FleetPilot

User Guide

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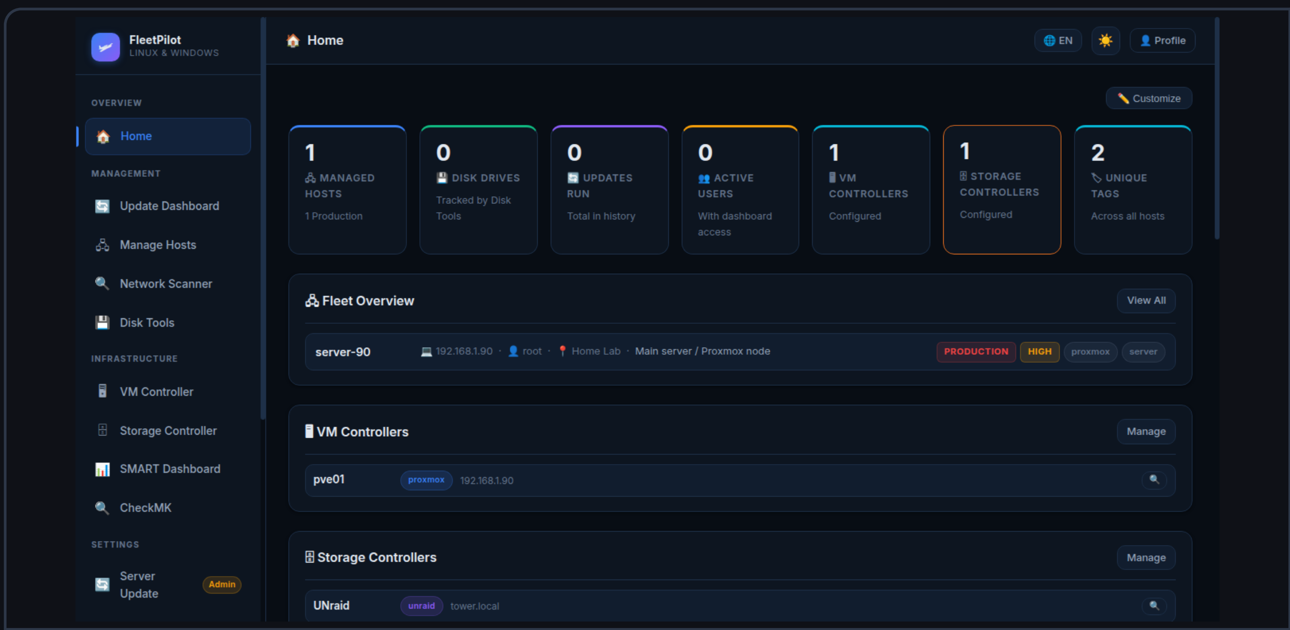

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01 Getting Started

FleetPilot is a centralized web platform for managing servers, VMs, storage systems, and disks in heterogeneous IT environments. Access is provided through any modern browser.

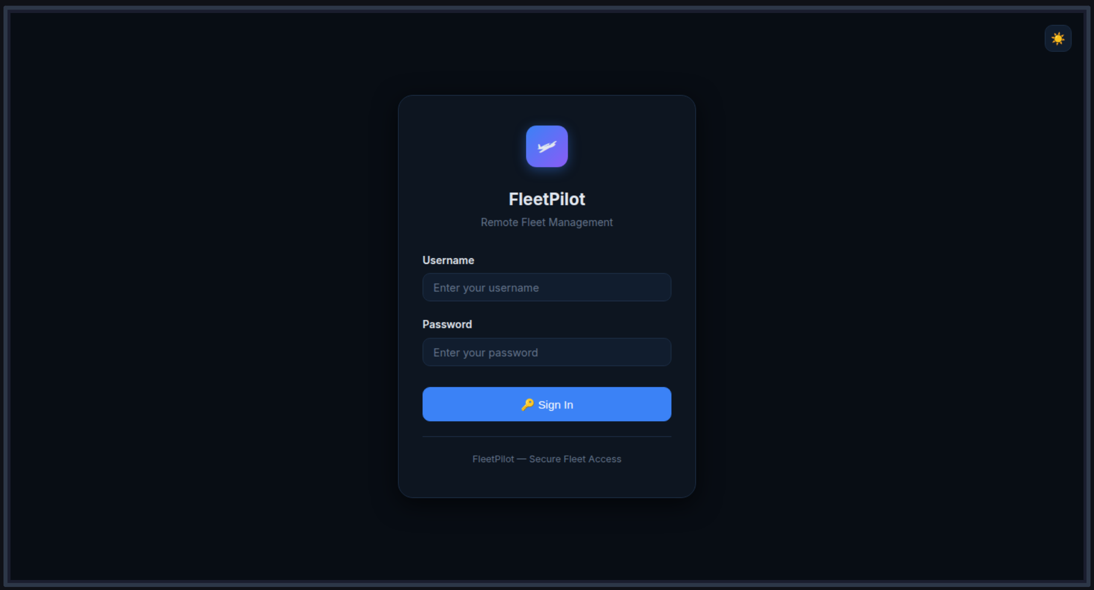

Figure 1.1 — FleetPilot Login Page

System Requirements

FleetPilot runs as a Python Flask application on Linux systems. A modern browser (Chrome, Firefox, Edge) is required for access. The application is optimized for screen resolutions of 1280×720 and above.

Logging In

Open the FleetPilot URL in your browser. Enter your username and password and click "Sign In". After successful authentication, you will be automatically redirected to the dashboard. The system includes built-in brute-force protection: after 10 failed attempts within 60 seconds, the IP address is temporarily blocked.

User Interface

The interface is divided into three areas: the left navigation bar with all modules, the main content area, and the header with language switcher, theme toggle, and profile menu. The dark mode design is optimized for extended working hours.

02 Dashboard

The dashboard is the central overview page of FleetPilot. It displays all key metrics at a glance and can be fully customized to your needs.

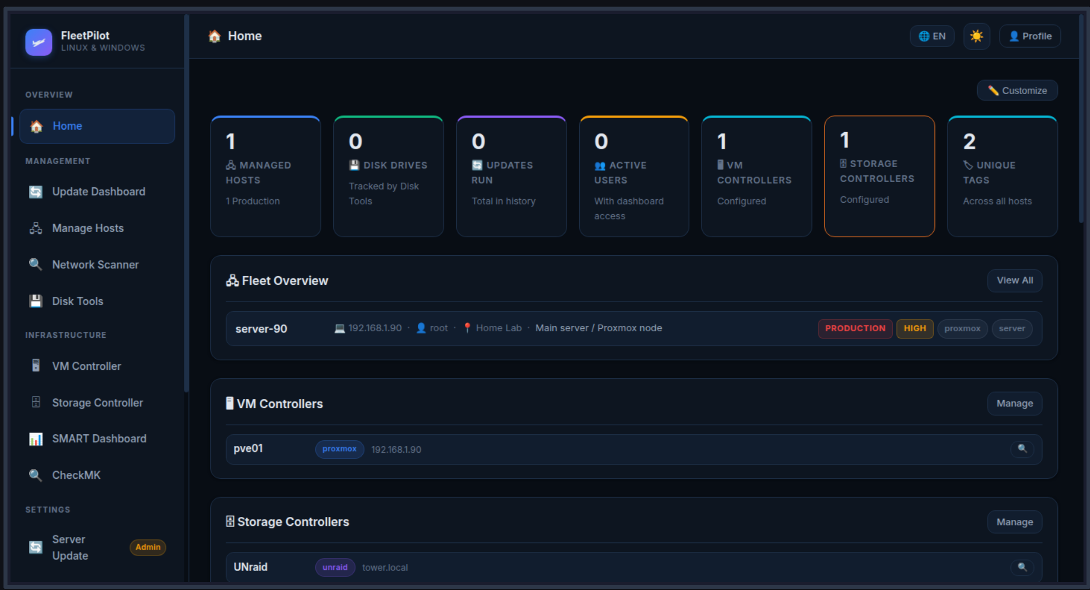

Figure 2.1 — Customizable FleetPilot Dashboard

Quick Stats

The seven metric cards at the top show: Managed Hosts, Disk Drives (via Disk Tools), Update Runs, Active Users, VM Controllers, Storage Controllers, and Unique Tags. Each card is color-coded and includes a brief description.

Customizing the Dashboard

Click "═■ Customize" in the top right to activate edit mode. In edit mode, you can drag and drop widgets into a new order. Click the red ✕ symbol on a widget to hide it. Hidden widgets appear in the "Hidden Widgets" panel and can be restored by clicking. Save the layout with "■ Save Layout". The layout is saved per user in the database.

Available Widgets

The dashboard offers 10 widgets: Quick Stats, Fleet Overview, VM Controllers, Storage Controllers, SMART Health Summary, Environment Breakdown, Tag Cloud, Recent Updates, Navigation Cards, and Plugin Widgets.

03 Host Management

Host management is the core of FleetPilot. All managed servers, PCs, and workstations are recorded and organized here.

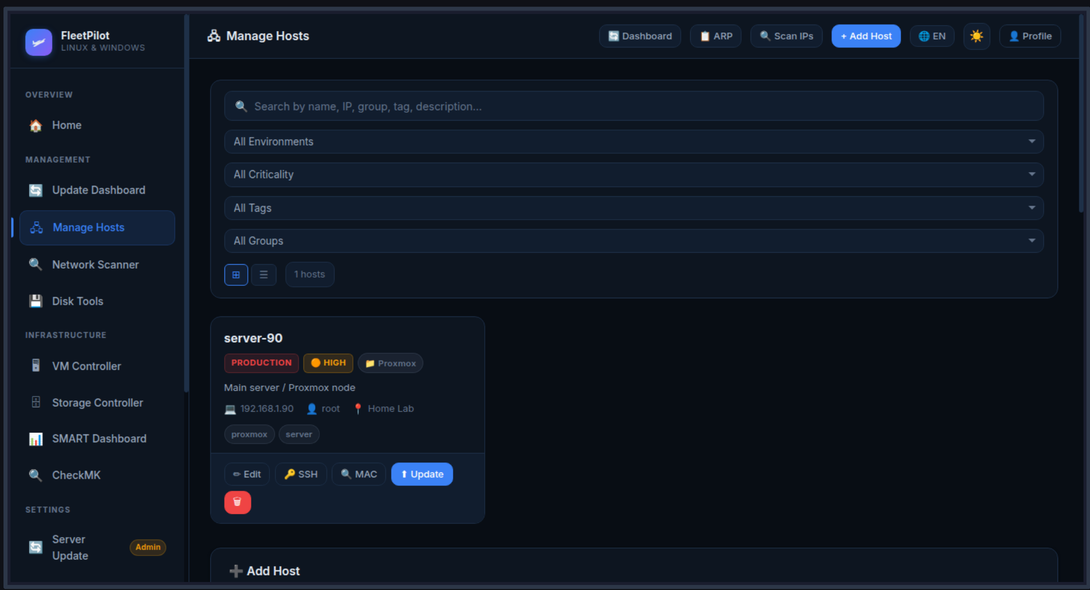

Figure 3.1 — Host Management with Filter Options

Adding a Host

Click "+ Add Host" in the header. Fill in the form: Name (required), IP address or hostname (validated for valid IPv4/IPv6 addresses and RFC-1123 hostnames), Environment (Production/Staging/Development/Testing), Criticality (Low/Medium/High/Critical), SSH user, group, tags, and description. Click "Save" to store.

Filtering and Searching Hosts

Use the search bar to filter hosts by name, IP, group, tag, or description. Dropdown menus allow filtering by environment, criticality, tags, and groups. The view can be toggled between card and list view.

Host Actions

Each host entry offers: Edit, SSH (direct SSH command), MAC (retrieve MAC address), Update (start system update), and Delete (red trash button). All actions are logged.

04 Update Manager

The Update Manager enables centralized management and execution of system updates on all managed hosts.

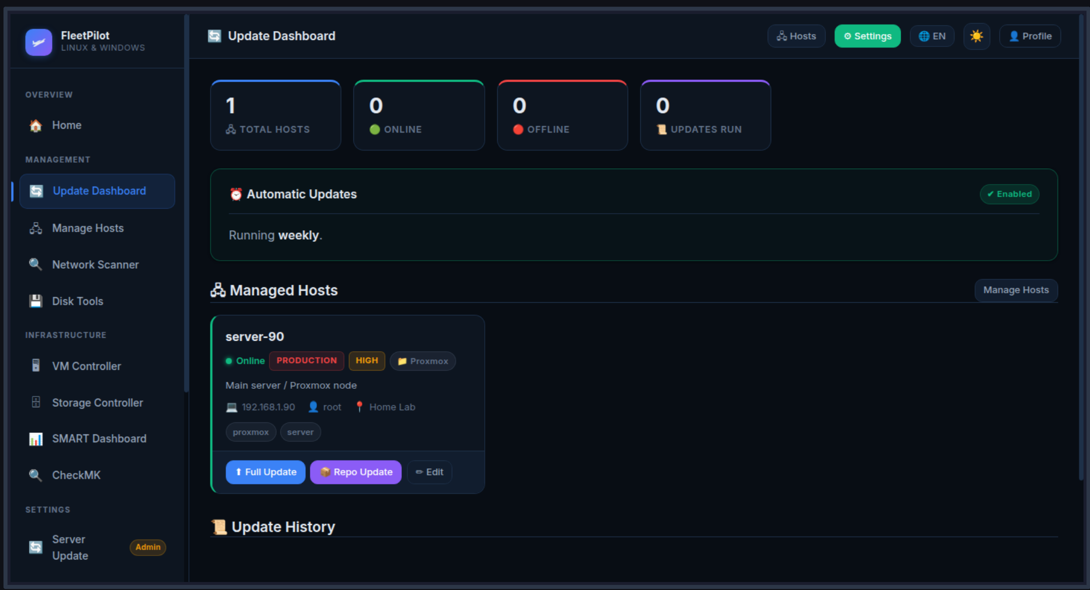

Figure 4.1 — Update Dashboard

Starting an Update

Navigate to a host and click "Update". Alternatively, manage all hosts at once via the Update Dashboard. The update process is displayed in real-time via Server-Sent Events (SSE) in the browser.

Update History

The Update Dashboard shows the history of all performed updates with timestamp, host name, and result (success/failure). The last 5 updates are also shown in the dashboard widget.

Automatic Updates

FleetPilot supports configuring automatic update schedules. Settings can be found in host properties under "Maintenance Window".

05 VM Controller

The VM Controller enables integration and monitoring of virtualization platforms such as Proxmox VE and Veeam.

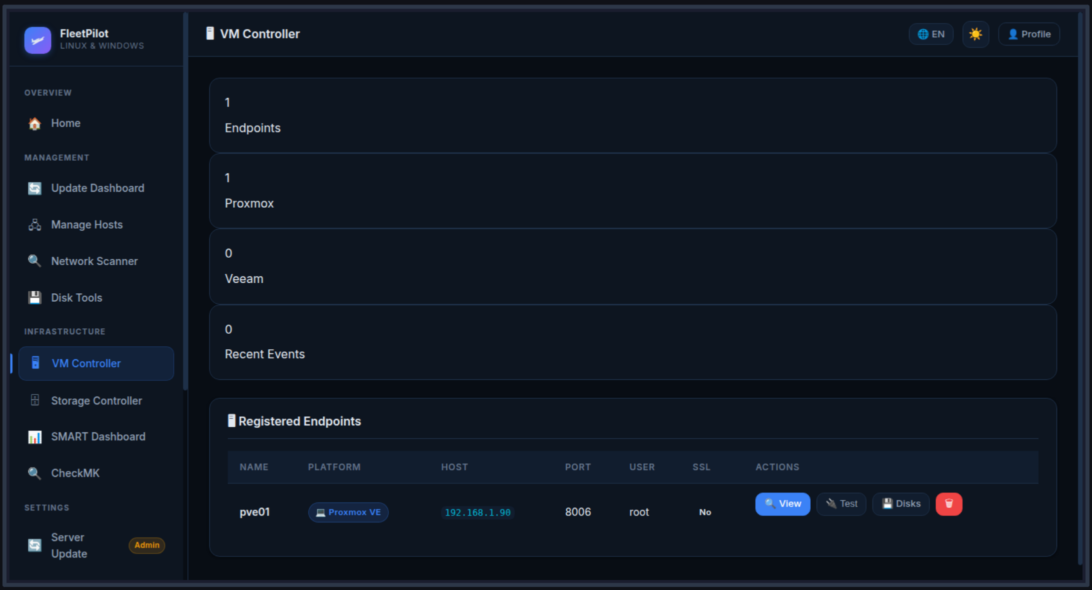

Figure 5.1 — VM Controller with Proxmox Integration

Connecting Proxmox VE

Click "+ Add Endpoint" and select "Proxmox VE". Enter host, port (default: 8006), user (default: root), and API token or password. SSL verification can be disabled for self-signed certificates. After saving, FleetPilot automatically tests the connection.

VM Overview

After a successful connection, all VMs and containers in the Proxmox cluster are listed. For each VM, status, CPU usage, RAM consumption, and uptime are displayed. The "View" button takes you directly to the Proxmox web interface.

Disk Inventory

The "Disks" button shows all physical hard drives of the Proxmox host with SMART status, model, capacity, and condition. This data is used for SMART monitoring.

06 Storage Controller

The Storage Controller integrates NAS and SAN systems such as TrueNAS, Synology, and other storage platforms.

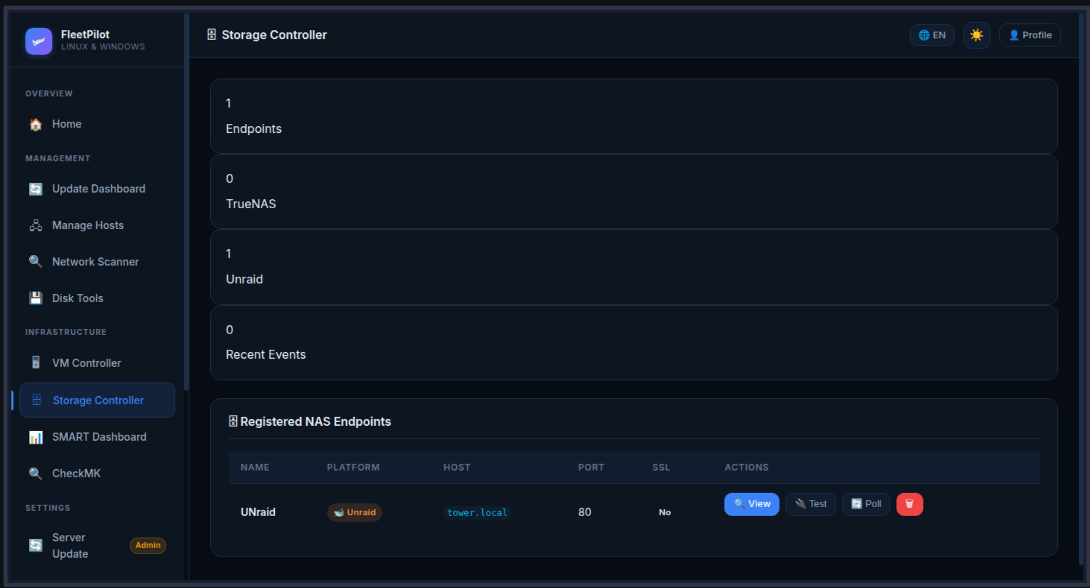

Figure 6.1 — Storage Controller Overview

Adding a Storage System

Click "+ Add Controller" and select the type (TrueNAS, Synology, Generic NAS). Enter the connection details. FleetPilot supports both HTTP and HTTPS with optional SSL verification.

Pool and Volume Overview

After connection, FleetPilot shows all storage pools, volumes, and their utilization. Critical fill levels are highlighted in color (Yellow: >75%, Red: >90%).

Disk Inventory

Similar to the VM Controller, physical hard drives of the storage system can be retrieved and integrated into SMART monitoring.

07 Disk Management & SMART

The Disk Management module monitors the health of all hard drives and SSDs via the SMART protocol.

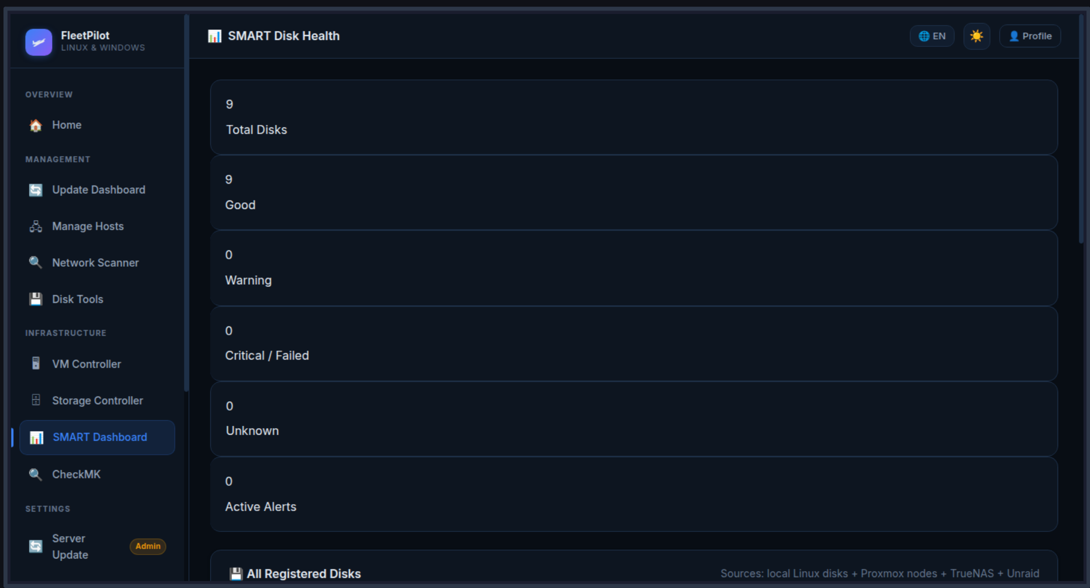

Figure 7.1 — SMART Dashboard

SMART Monitoring

The SMART dashboard shows all monitored hard drives with their current health status: Healthy (green), Warning (yellow), or Critical (red). Critical SMART attributes such as Reallocated Sectors, Pending Sectors, and Uncorrectable Errors are highlighted.

Disk Tools

The Disk Tools module enables direct disk management: start SMART test, display detailed attribute list, export disk inventory. All actions are logged.

Automatic Polling

FleetPilot polls SMART data automatically at configurable intervals. A manual poll can be triggered via the "Poll Now" button. Critical status changes are displayed in the dashboard widget.

08 CheckMK Integration

FleetPilot integrates seamlessly with CheckMK for advanced monitoring functions.

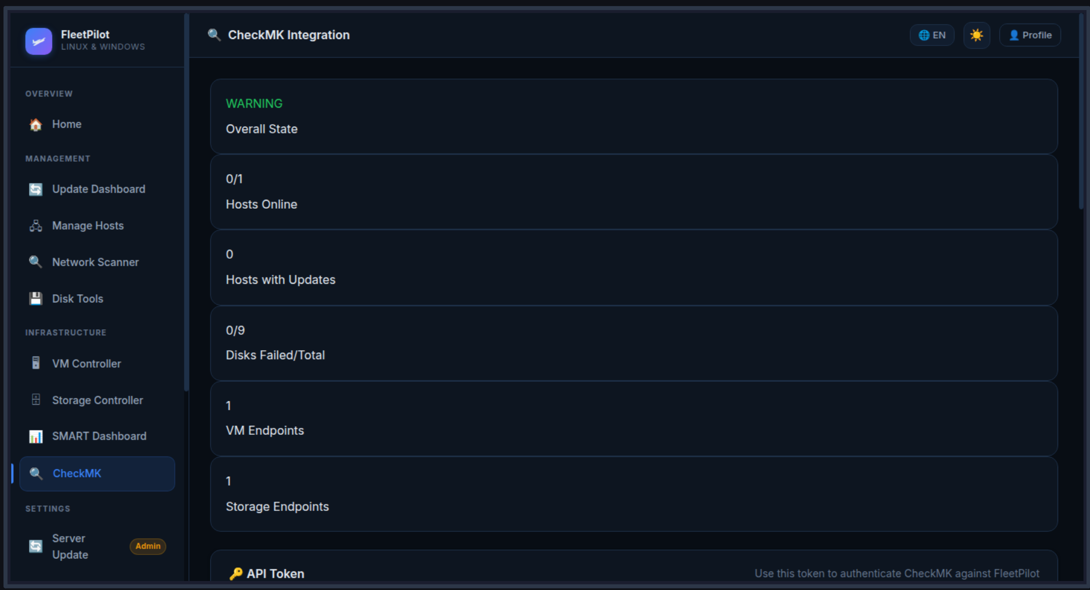

Figure 8.1 — CheckMK Integration Dashboard

Connecting CheckMK

Navigate to "CheckMK" in the sidebar. Enter the CheckMK URL, site name, and an API token. The token must be created in CheckMK under "Users → API Tokens". Click "Test Connection" to verify.

API Authentication

All CheckMK API requests require a valid Bearer token. This is transmitted in the HTTP header as "Authorization: Bearer ". The token status can be retrieved via "/api/checkmk/token_info".

Monitoring Data

After a successful connection, FleetPilot displays host status, service states, and active alerts from CheckMK. Data is updated in real-time.

09 User Management

FleetPilot has a complete user management system with role-based access control.

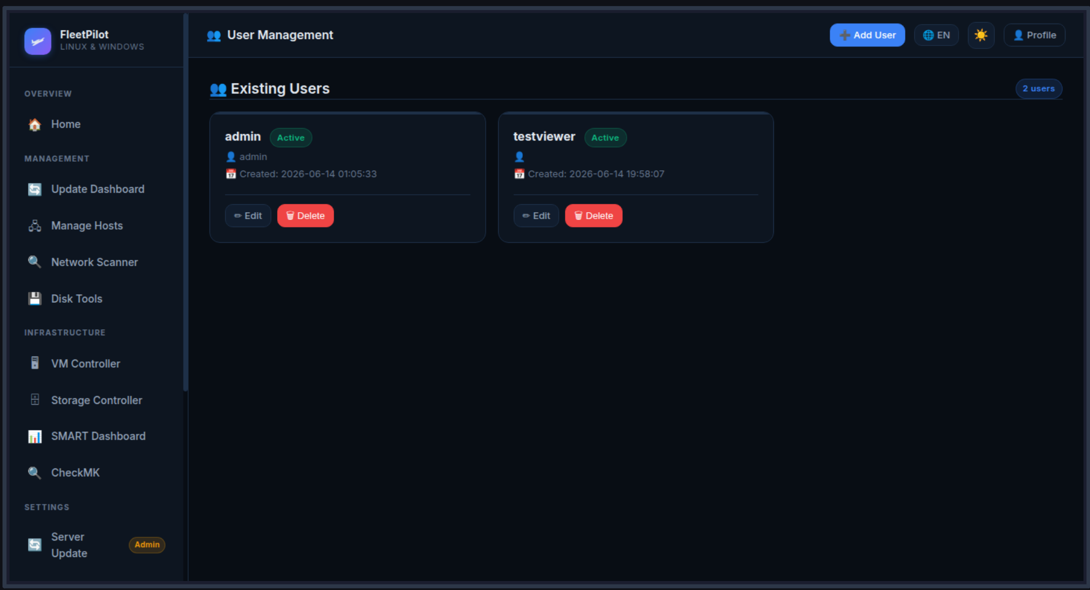

Figure 9.1 — User Management

Creating Users

Only administrators can create new users. Navigate to "Server Update → Users" and click "+ Add User". Specify username, email, password, and role. Available roles: Admin (full access), Operator (read + actions), Viewer (read only).

Password Policies

Passwords must be at least 8 characters long. The system uses scrypt as the hashing algorithm for maximum security. Passwords can only be changed by the user themselves or an administrator.

Dashboard Layouts

Each user has their own dashboard layout saved in the database. Layouts can be reset via "Reset to Default". Administrators cannot view user layouts.

10 Plugins & Extensions

FleetPilot supports a plugin system for custom extensions and integrations.

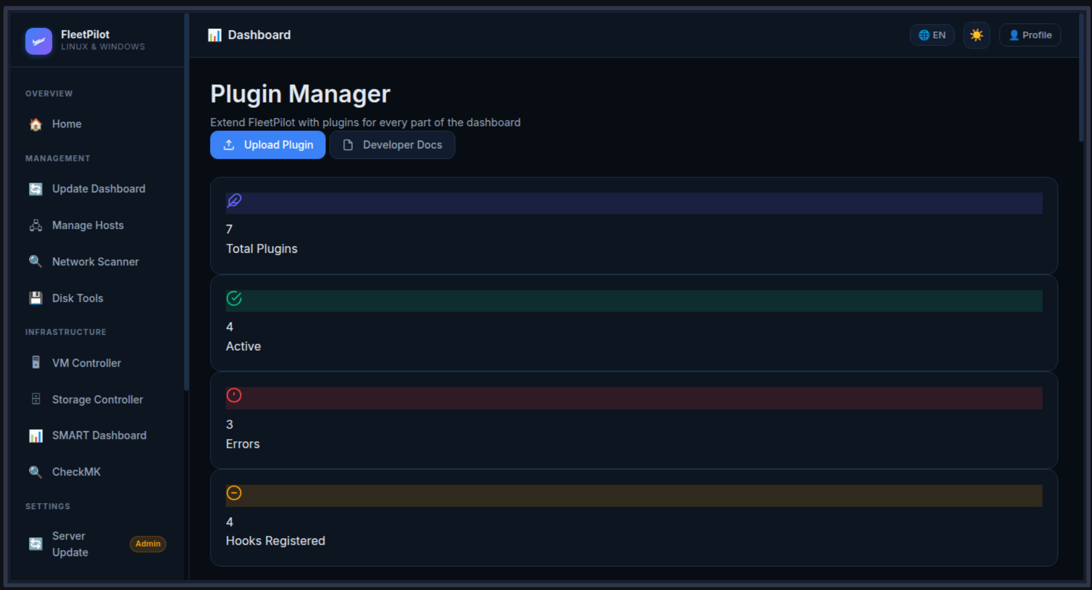

Figure 10.1 — Plugin Manager

Installing Plugins

Navigate to "Plugins" in the sidebar. Available plugins are loaded from the plugin repository. Click "Install" next to a plugin to install it. After installation, the plugin is immediately active.

Plugin Development

Custom plugins can be developed as Python modules. The plugin interface is documented and allows integration of custom monitoring sources, actions, and dashboard widgets.

Network Scanner

The built-in Network Scanner (module "Scanner") enables automatic device discovery on the network via ARP scan and IP ping sweep. Found devices can be added directly as hosts.